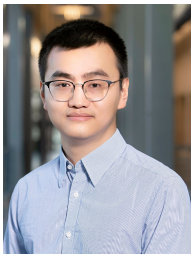
Jilin Hu

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Education and Employment

- 2023 – present: **Professor, Zijiang Young Scholar**, School of Data Science and Engineering, East China Normal University.
- 2022 – 2023: **Associate Professor**, Department of Computer Science, Aalborg University, Denmark.
- 2020 – 2022: **Assistant Professor**, Department of Computer Science, Aalborg University, Denmark.
- 2019 – 2020: **Research Scientist**, Inception Institute of Artificial Intelligence, United Arab Emirates.
- 2017 – 2018: **Visiting PhD Student**, Intelligent Transportation Research Group, University of California, Berkeley, USA.
- 2015 – 2019: **PhD Student**, Department of Computer Science, Aalborg University, Denmark.
- 2011 – 2014: **MSc Student**, School of Aeronautics and Astronautics, Shanghai Jiao Tong University, China.
- 2007 – 2011: **BSc Student**, School of Metrology and Testing Engineering, China Jiliang University, China.

Research Grants

- 2026.08 – 2028.08: **Ministry of Science and Technology**, National Science and Technology Major Project for New Generation Artificial Intelligence, Principal Investigator, Hypothesis-driven Intelligent Design of Complex Chemical Formulations, RMB 11.748M (including RMB 3.748M in central government funding).
- 2025 – 2028: **National Natural Science Foundation of China**, General Program, Principal Investigator, Research on Multimodal Universal Pretraining for Trajectory Analytics, RMB 500k.
- 2023 – 2026: **National Natural Science Foundation of China**, Excellent Young Scientists Fund (Overseas), Principal Investigator, Spatio-temporal Data Management and Analytics, RMB 1M.
- 2024 – 2026: **Villum Foundation**, Synergy, Co-Principal Investigator, Inverse Design of Materials Using Diffusion Probabilistic Models, DKK 2.99M.
- 2023 – 2026: **East China Normal University Basic Research Funds**, Key Research Initiative, Principal Investigator, Personalized Deposit Rate Pricing for Banking Customers, RMB 500k.

Honors and Awards

- 2025: **EMNLP 2025 Best Paper (Industrial) Award**.
- 2024: **VLDB 2024 Best Paper Award Nomination**.
- 2023: **ACM SIGSPATIAL China Rising Star Award**.

Honors and Awards (continued)

2022
- **ICDE 2022 Best Paper Award**.
- **Selected for the National Youth Talent Program**.

Professional Service

2026 – present
- **Member, CCF Big Data Expert Committee**.

2024 – present
- **Member, CCF Large Model Forum**.
- **Member, Intelligent and Digital Surgery Committee, Shanghai Medical Association**.

2023 – present
- **Member, CCF Database Technical Committee**.
- **Member, CAAI Intelligent Service Technical Committee**.

2022 – present
- **Chair, Denmark Chapter, IEEE Computer Society**.
- **Member, ACM SIGSPATIAL China Chapter**.

Publications

Journal Articles

[1] Jonas A. Finkler, Yan Lin, Tao Du, **Jilin Hu**, and Morten M. Smedskjaer. "Inverse Design of Amorphous Materials With Targeted Properties". In: *Advanced Materials* (2026), e22493.

[2] Shiyan Hu, Kai Zhao, Chenjuan Guo, Xiangfei Qiu, Yang Shu, **Jilin Hu**, Christian S. Jensen, and Bin Yang. "TAP: Time Series Anomaly Prediction Via Adaptive Period Modeling and Dual Representation Learning". In: *IEEE Transactions on Knowledge and Data Engineering* (2026), pp. 1–14.

[3] Sean Bin Yang, Ying Sun, Hao Miao, **Jilin Hu**, Bin Yang, Torben Bach Pedersen, and Christian S. Jensen. "AutoPRL: Automated Path Representation Learning via Learnable View Generator". In: *IEEE Transactions on Knowledge and Data Engineering* (2026).

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[5] Haomin Yu, **Jilin Hu**, Kasper Tolborg, Chenjuan Guo, Mohamed Saraee, Bin Yang, and Christian S. Jensen. "A Selective Multi-View Representation Augmentation Framework for Crystal Property Prediction". In: *ACM Transactions on AI for Science* (2026).

[6] Kai Zhang, Chaoxiang Lan, Yazhang Xu, Zheyang Li, Wenming Tan, Ye Ren, and **Jilin Hu**. "SparseLight: Dynamic gradient-optimized softmax for efficient transformer acceleration". In: *Knowledge-Based Systems* 333 (2026), p. 115050.

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[10] Di Qi, **Jilin Hu**, and Jianbing Shen. "Few-Shot Object Detection With Self-Supervising and Cooperative Classifier". In: *IEEE Transactions on Neural Networks and Learning Systems* 35.4 (2024), pp. 5435–5446.

[11] Xinle Wu, Xingjian Wu, Bin Yang, Lekui Zhou, Chenjuan Guo, Xiangfei Qiu, **Jilin Hu**, Zhenli Sheng, and Christian S. Jensen. "AutoCTS++: zero-shot joint neural architecture and hyperparameter search for correlated time series forecasting". In: *The VLDB Journal* 33.5 (2024), pp. 1743–1770.

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[13] Simin Xu, Lingkun Luo, Haichao Hong, **Jilin Hu**, Bin Yang, and Shiqiang Hu. "Multi-granularity attention in attention for person re-identification in aerial images". In: *The Visual Computer* 40.6 (2024), pp. 4149–4166.

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[15] Simin Xu, Lingkun Luo, **Jilin Hu**, Bin Yang, and Shiqiang Hu. "Semantic driven attention network with attribute learning for unsupervised person re-identification". In: *Knowledge-Based Systems* 252 (2022), p. 109354.

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[23] **Jilin Hu**, Bin Yang, Christian S. Jensen, and Yu Ma. "Enabling time-dependent uncertain eco-weights for road networks". In: *GeoInformatica* 21.1 (2017), pp. 57–88.

Conference Papers

[1] Yujie Li, Zezhi Shao, Chengqing Yu, Yisong Fu, Weijie Zhu, Yifan Du, **Jilin Hu**, Bin Yang, Yongjun Xu, and Fei Wang. "QUALS: Corpus Equilibrium for Universal Forecasting via Pattern Quantization and Learnability Synchronization". In: *PVLDB*. 2027.

[2] Chaofan Wang, Ziqiang Yu, Xiaohui Yu, **Jilin Hu**, Meng Chen, and Yang Liu. "CRSTNet: Efficient Spatio-Temporal Forecasting via Adaptive Cluster Routing". In: *ICDE*. 2027.

[3] Sean Bin Yang, Ying Sun, Zongyi Xu, Tung Kieu, **Jilin Hu**, Bin Yang, Kristian Torp, Hua Lu, and Torben Bach Pedersen. "When Correlations Mislead: Confounder-Aware Multi-View Urban Region Representation Learning". In: *ICDE*. 2027.

[4] Sean Bin Yang, Zongyi Xu, **Jilin Hu**, Ying Sun, Kristian Torp, Bin Yang, Torben Bach Pedersen, and Christian S. Jensen. "Rethinking Multi-View Fusion for Urban Representation Learning". In: *PVLDB*. 2027.

[5] Yuhan Zhu, **Jilin Hu**, Xinying Cai, Yingshan Li, Li Ma, Xiangfei Qiu, Linsen Li, Kai Zhang, Yao Fu, Weihao Jiang, and Bin Yang. "WPBench: A Comprehensive Benchmark for Wind Power Forecasting". In: *ICDE*. 2027.

[6] Yongru Chen, Kai Zhang, Zeliang Zong, Yuchen Lu, Wenming Tan, Ye Ren, and **Jilin Hu**. "One Layer's Trash is Another Layer's Treasure: Adaptive Layer-wise Visual Token Selection in LVLMs". In: *CVPR*. 2026.

[7] Hongfan Gao, Wangmeng Shen, Bin Yang, and **Jilin Hu**. "HyFAD: Hybrid Time-Frequency Diffusion with Frequency-Aware Embedding for Time Series Imputation". In: *NeuriPS*. 2026.

[8] Chenxu Han, Sean Bin Yang, and **Jilin Hu**. "DiffMM: Efficient Method for Accurate Noisy and Sparse Trajectory Map Matching via One Step Diffusion". In: *AAAI*. 2026.

[9] Zhengyu Li, Xiangfei Qiu, Yuhan Zhu, Xingjian Wu, **Jilin Hu**, Chenjuan Guo, and Bin Yang. "GCGNet: Graph-Consistent Generative Network for Time Series Forecasting with Exogenous Variables". In: *ICLR*. 2026.

[10] Yan Lin, **Jilin Hu**, Shengnan Guo, Christian S. Jensen, Youfang Lin, and Huaiyu Wan. "DiSGMM: A Method for Time-varying Microscopic Weight Completion on Road Networks". In: *IJCAI*. 2026.

[11] Yan Lin, **Jilin Hu**, N. M. Anoop Krishnan, and Morten M. Smedskjaer. "AMGenC: Generating Charge Balanced Amorphous Materials". In: *NeuriPS*. 2026.

[12] Caini Liu, Shuhui Gong, Yi Xu, **Jilin Hu**, and Hongbin Pei. "Adaptive Road Semantic Self-Learning for Kilo-metre level Vehicle Trajectory Prediction". In: *SIGSPATIAL*. Short Paper. 2026.

[13] Xvyuan Liu, Xiangfei Qiu, Hanyin Cheng, Xingjian Wu, Chenjuan Guo, Bin Yang, and **Jilin Hu**. "ASTGI: Adaptive Spatio-Temporal Graph Interactions for Irregular Multivariate Time Series Forecasting". In: *ICLR*. 2026.

[14] Xvyuan Liu, Xiangfei Qiu, Xingjian Wu, Zhengyu Li, Chenjuan Guo, **Jilin Hu**, and Bin Yang. "Rethinking Irregular Time Series Forecasting: A Simple yet Effective Baseline". In: *AAAI*. 2026.

[15] Jiahong Lyu, Hongfan Gao, Wangmeng Shen, Qingsong Zhong, and **Jilin Hu**. "PolarFormer: Radial–Angular Latent Modeling for Unconditional Time Series Generation". In: *KDD*. 2026.

[16] Xiaowei Mao, Huihu Ding, Yan Lin, Tingrui Wu, Shengnan Guo, Dazhuo Qiu, Feiling Fang, **Jilin Hu**, and Huaiyu Wan. "Spatial-Temporal Feedback Diffusion Guidance for Controlled Traffic Imputation". In: *AAAI*. 2026.

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[22] Wangmeng Shen, Hongfan Gao, Qingsong Zhong, Dingli Xu, and **Jilin Hu**. "FSDI: Frequency-Shaped Diffusion For Time-Series Imputation". In: *The Web Conference*. 2026.

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[24] Xingjian Wu, Xvyuan Liu, Junkai Lu, Siyuan Wang, Xiangfei Qiu, Yang Shu, **Jilin Hu**, Chenjuan Guo, and Bin Yang. "ST-EVO: Towards Generative Spatio-Temporal Evolution of Multi-Agent Communication Topologies". In: *EMNLP*. 2026.

[25] Xiangyu Xu, Qingsong Zhong, and **Jilin Hu**. "TimeMar: Multi-Scale Autoregressive Modeling for Unconditional Time Series Generation". In: *The Web Conference*. 2026.

[26] Yi Xu, Shuhui Gong, Jia Wang, Tong Bian, Yuchen Yang, **Jilin Hu**, and Hongbin Pei. "Graceful Degradation in Trajectory User Linking: A Regime-Adaptive Aggregation Framework". In: *SIGSPATIAL*. 2026.

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[29] Xiaoyun Yu, Li Fan, Xiangfei Qiu, Nanqing Dong, Yonggui Huang, Honggang Qi, Geguang Pu, Wanli Ouyang, Xi Chen, and **Jilin Hu**. "MEMTS: Internalizing Domain Knowledge via Parameterized Memory for Retrieval-Free Domain Adaptation of Time Series Foundation Models". In: *KDD*. 2026.

[30] Buang Zhang, Tung Kieu, Xiangfei Qiu, Chenjuan Guo, **Jilin Hu**, Aoying Zhou, Christian S. Jensen, and Bin Yang. "An Encode-then-Decompose Approach to Unsupervised Time Series Anomaly Detection on Contaminated Training Data". In: *ICDE*. 2026.

[31] Qingsong Zhong, Haomin Yu, Yan Lin, Wangmeng Shen, Long Zeng, and **Jilin Hu**. "SculptDrug: A Spatial Condition-Aware Bayesian Flow Model for Structure-based Drug Design". In: *AAAI*. 2026.

[32] Zeliang Zong, Kai Zhang, Yarong Wang, Wenming Tan, Ye Ren, and **Jilin Hu**. "Less Token, More Signal: MoE Expert Pruning via Critical Token Selection". In: *ICML*. 2026.

[33] Hongfan Gao, Wangmeng Shen, Xiangfei Qiu, Ronghui Xu, Bin Yang, and **Jilin Hu**. "SSD-TS: Exploring the potential of linear state space models for diffusion models in time series imputation". In: *KDD*. 2025.

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[36] Xiangfei Qiu, Zhe Li, Wanghui Qiu, Shiyan Hu, Lekui Zhou, Xingjian Wu, Zhengyu Li, Chenjuan Guo, Aoying Zhou, Zhenli Sheng, **Jilin Hu**, Christian S. Jensen, and Bin Yang. "TAB: Unified Benchmarking of Time Series Anomaly Detection Methods". In: *PVLDB*. 2025.

[37] Xiangfei Qiu, Xingjian Wu, Hanyin Cheng, Xvyuan Liu, Chenjuan Guo, **Jilin Hu**, and Bin Yang. "DBLoss: Decomposition-based Loss Function for Time Series Forecasting". In: *NeuriPS*. 2025.

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[39] Haichen Wang, Liu Yang, Xinyuan Zhang, Haomin Yu, Ming Li, and **Jilin Hu**. "ADFormer: Aggregation Differential Transformer for Passenger Demand Forecasting". In: *IJCAI*. 2025.

[40] Tonglong Wei, Yan Lin, Youfang Lin, Shengnan Guo, **Jilin Hu**, Haitao Yuan, Gao Cong, and Huaiyu Wan. "PLMTrajRec: A Scalable and Generalizable Trajectory Recovery Method with Pre-trained Language Models". In: *NeuriPS*. 2025.

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[42] Yongfu Wei, Yan Lin, Hongfan Gao, Ronghui Xu, Sean Bin Yang, and **Jilin Hu**. "Path-LLM: A Multi-Modal Path Representation Learning by Aligning and Fusing with Large Language Models". In: *WWW*. Ed. by Guodong Long, Michale Blumestein, Yi Chang, Liane Lewin-Eytan, Zi Helen Huang, and Elad Yom-Tov. ACM, 2025, pp. 2289–2298.

[43] Wenyu Wu, Jiali Mao, Jiafan Liu, Yixiao Tong, Lisheng Zhao, Shaosheng Cao, **Jilin Hu**, Aoying Zhou, and Lin Zhou. "CDMap: Complementarity and Disparity-aware Map Inference Quality Enhancement". In: *ICDE*. 2025.

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[47] Ziyan Yang, Shuhui Gong, Xinqi Liu, Jiahao Lv, Changjian Liu, **Jilin Hu**, and Hongbin Pei. "VisitFrequency-Diffusion: Leveraging Recurrent Visits for Long-Term Individual Trajectory Forecasting". In: *SIGSPATIAL*. 2025, pp. 1190–1193.

[48] Yi Zhang, Kai Zhang, Zheyang Li, Wenming Tan, Ye Ren, and **Jilin Hu**. "Beyond Dynamic Quantization: An Efficient Static Hierarchical Mix-precision Framework for Near-Lossless LLM Compression". In: *EMNLP Industry*. 2025.

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[50] Zeliang Zong, Kai Zhang, Zheyang Li, Wenming Tan, Ye Ren, yiyan zhai, and **Jilin Hu**. "1+1>2: A Synergistic Sparse and Low-Rank Compression Method for Large Language Models". In: *EMNLP Findings*. 2025.

[51] Christian S. Jensen, Bin Yang, Chenjuan Guo, **Jilin Hu**, and Kristian Torp. "Routing with Massive Trajectory Data". In: *ICDE*. 2024, pp. 5542–5547.

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